INCH-POUND

MIL-P-70679 (AR)
AMENDMENT 3
14 September 1995

SUPERSEDING
AMENDMENT 2
15 March 1991

MILITARY SPECIFICATION

PROJECTILE, 60MM, WP, XM722 (M722) BODY ASSEMBLY

This amendment forms a part of MIL-P-70679 (AR), dated 14 January 1988, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1: Delete "MIL-P-63194 . . . Charge" in its entirety.

PAGE 2

* 2.1.1: Delete "MIL-STD-810 . . . Guidelines" in its entirety.

* 2.2, Under "E10 . . . Materials: Add : "B117 - Standard Test Method of Salt Spray (Fog) Testing."

PAGE 4

* 3.8: Delete in its entirety and substitute:

"3.8 Hydrotest. Each body shall successfully withstand the hydrostatic test load specified on dwg. 9392366 when tested in accordance with 4.4.3.3 and 4.4.5.5."

AMSC N/A FSC 1310
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

* 3.9: Delete in its entirety and substitute:

"3.9 Providing ground tests.

3.9.1 Maximum average pressure. Projectile bodies shall reveal no evidence of failure when tested in accordance with 4.4.5.6.1. Failure shall be:

a. Projectile body deformation in excess of .006 inch attributable to noncompliance with drawing or specification requirements.

b. Loss or breakup of any component in the mortars tube or flight.

3.9.2 Service pressure. Projectile bodies shall reveal no evidence of failure when tested at service pressure in accordance with 4.4.5.6.2. Failure shall be:

a. Projectile body deformation in excess of .006 inch attributable to noncompliance with drawing or specification requirements .

b. Loss or breakup of any component in the mortars tube or flight .

c. A round falling less than 80 percent of the average range of the other projectiles fired at service pressure.

PAGE 5

* 4.3.1, After "Body prior to obturating band groove: Add : "and press-fit diameter".

Delete: "... before paint" and substitute: "after final machining".

PAGE 6

* Table I, After "Body, prior to machining obturating band groove": Add: "and press-fit diameter".

Delete: ".. "Hydrotest . . . 4.4.5.5" in its entirety and substitute "Examination for defects ... 10 . . . (-) . . . 3.2 . . . 4.4.2.1".

After second heading ("Body") : Add : "after final machining".

Delete: "Maximum average pressure test" and substitute: "Proving ground tests".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

PAGE 7

* 4.4.2: Delete "Note . . . found." in its entirety and substitute:

"NOTE: Tool control inspection. Dimensions marked tool control shall be gaged during First Article inspection at the beginning of each workday, every four (4) hours during production, for the last component prior to and first component after each tooling, equipment, or process change, repair or adjustment. Measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimension prior to substitution. When acceptability has been determined inspection of tool dimensions will be permitted. Whenever a defective tool control dimension is found immediate action will be taken by the contractor to correct the tool problem and all material produced since the last successful inspection shall be rejected."

PAGE 8

* 4.4.2.1: Add Majors as follows:

"101 Wall thickness at rear taper (.125 - .030) . . . (-) . . . 0.40% . . . 3.2 Gage

"102 Wall thickness at bourrelet . . . (-) . . . 40% . . . 3.2 Gage

"103 Wall thickness at rear of obturator groove location, min. . . . (-) . . . 0.40% . . . 3.2 Gage".

PAGE 9

* 4.4.2.2, After title "Body" Add : "after Final Machining".

Major 108: Delete in its entirety.

PAGE 10

* 4.4.2.2, Major 109: Delete in its entirety.

Major 119: Delete in its entirety.

Major 123: Delete "Profile . . . incorrect" and substitute "Outside diameter at 5 basic lengths".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

PAGE 12

4.4.2.3, Major 101: Change AQL "100%" to read ".40%".

* 4.4.2.3, Major 102: Delete "phosphate" and substitute “wash primer".

Note 1: Delete "4.4.5.6" and substitute "4.4.3.4".

PAGE 16

* 4.4.3.3: Add the following:

"A temporary internal thread shall be machined in the forward end of the body to couple the body to the hydrostatic test fixture. Only the threads and front face of the projectile body may be in contact with the test fixture. After the required minimum pressure is met or exceeded the pressure shall be held for five (5) seconds, minimum. Projectile rupture, cracking, leakage or bulging during the test shall be cause for rejection of the body.

If more than one projectile is tested simultaneously on a manifold and one projectile fails the test it shall be replaced with a new body. All other projectiles attached to the manifold at that time shall be retested with the new body until all simultaneously meet the time and pressure requirements."

PAGE 17

* 4.4.3.4: Delete text above Table II in its entirety and substitute:

"4.4.3.4 Proving ground test. (See 4.4.5.6).

4.4.3.4.1 Maximum average pressure test. Five (5) projectiles shall be tested from the first article sample, from each of the first five (5) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. Defect classification and individual accept/reject criteria shall be in accordance with Table II and 3.9.

* 4.4.3.4.2: Service pressure test. Sampling and accept/reject criteria shall be as specified in 4 .4.3.4.1 except that the sample size shall be twenty (20) projectiles."

* 4.4.5.1.1 a: Delete "sufficient" and substitute "depth sufficient to remove the".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

* 4.4.5.4: Delete in its entirety and substitute:

"4.4.5.4 Paint adhesion/salt spray tests. In performing the following tests, TT-C-490 shall apply unless specified otherwise, herein. The tests shall be performed for first article inspection, each day of painting during production and whenever a change is made to the paint process, until all samples from five (5) consecutive days of painting have passed the tests. Thereafter, the sample frequency may be reduced to twice per week during continuous days of painting.

The specimens shall consist of painted body assemblies, randomly selected from assemblies painted at the same time and in the same equipment as used in regular production. The samples shall be taken from each continuous final paint operation, up to 24 hours long. One projectile shall be taken during the first thirty (30) minutes of painting, and the others at any other time.

The paint adhesion test shall be performed on each projectile prior to salt spray testing using the procedures and pass/fail criteria contained in 4.2.8 and 3.5.6 respectively, in TT-C-490. After passing the adhesion test each specimen shall be scored lengthwise using a precision scribing tool/insert or equivalent in preparation for the salt spray test.

The duration of the salt spray test shall be as specified on drawing 9395680. The salt solution, test equipment, and drying procedures shall be in accordance with ASTM B117, Salt Spray (Fog) Testing, unless specified otherwise, herein. Specimens shall be positioned between 15 and 30 degrees from vertical during the test. No coating of any kind shall be permitted over the paint on stamp marks, however, areas in contact with racks or supports, unpainted surfaces and areas within 1/8 inch of unpainted surfaces may be protected. The significant surface shall be specified in 4.2.9 of TT-C-490. Although the failure criteria in a-g below applies to the significant surface, the government may also reject a specimen for variations of sections a-g, or for an unlisted defect(s) anywhere on the part, if deemed excessive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

PAGE 19 (cont)

As used herein, the term "blister" refers to an unbroken area of paint with a bubble-like appearance, whereas "spots" are exposed phosphate pretreatment, bare metal, or corrosion (rust) spots breaking through the paint. Blisters which are present when the test is over are to be counted within five (5) minutes of removal from the test chamber. After testing, any projectile which fails for any of the following reasons shall be cause for rejection of all parts painted since the last successful test and a return to the original sampling plan if failure occurred during reduced sampling phase.

a. The paint shows more than 1/8 inch creepage, blistering or loss of adhesion of paint from the scribe mark, stamp mark, or edge.

b. There is more than one (1) blister.

c. Any blister exceeds 3/64 inch average diameter.

d. Any spot exceeds 3/32 inch average diameter.

e. There are two (2) or more scattered pits exceeding 1/16 inch average diameter.

f. There are four (4) or more spots, total.

g. There is evidence of rust under the paint (a brown discoloration showing through the paint) in excess of 1/4 inch in diameter, total area."

PAGE 20

4.4.5.5, sentence 7 (The test . . . pressure."): Delete in its entirety.

After the last sentence add:

"Examination should be made of the hydrotest plug after a projectile is blown during testing. Badly nicked or damage plugs may cause destruction of acceptable items. The protective barrier must allow sufficient clearance for a defective projectile to completely fail in any direction."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

* 4.4.5.6: Delete in its entirety and substitute:

"4.5.6 Proving ground tests

"4.5.6.1 Maximum average pressure test. (See 4.4.3.4.1) The samples supplied in accordance with Table I and 4.4.2.3 (inert loaded to the service weight and dummy fuzed) shall be assembled into complete rounds. This test shall be fired at charge 4 from a horizontally mounted mortar into cellotex and propelling charges adjusted or supplemented to produce 9363 to 10,050 psi at 70 degrees F. Smear camera equipment shall record the flight of the projectile. The test shall be performed at a government proving ground in accordance with applicable U.S. Army TECOM acceptance test procedures. Cartridges shall be conditioned at 70 degrees plus or minus 2.5 degrees F for 24 hours prior to firing. All rounds shall be recovered.

4.5.6.2 Service pressure test. (See 4.4.3.4.2) The sample supplied in accordance with Table I and 4.4.2.3 (inert loaded to the service weight and dummy fuzed) shall be assembled into complete rounds. This test shall be fired at charge 4 from a mortar for which the round is standard, the tube of which has a minimum of 25 percent life remaining based on current mortar tube condemnation criteria. Radar and smear camera equipment shall record the flight of the projectile to help determine compliance with 3.9.2. The test shall be performed at a government proving ground in accordance with applicable U.S. Army TECOM acceptance test procedures. Cartridges shall be conditioned at 70 degrees plus or minus 2.5 degrees F for 24 hours prior to firing. Provisions shall be made to recover any projectile fired in the test which falls short (less than 80 percent of average (see 3.9.2c)) or exhibits abnormal behavior.

4.5.6.3 Test anomalies. Should a premature burst or metal parts separation occur, a photograph shall be taken of the evidence and attached to the firing record. The velocity and range standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day, weather and the time of firing shall also be recorded.

HAWK radar or equivalent shall be used to observe for metal parts separation and unobserved short rounds. In these events and analysis should be made of the radar data and should be noted on the firing record. This shall include a comparison of times of flight of the unobserved rounds with that of the normal rounds, and prints of the preceding and succeeding rounds, shall be attached to the firing record. Data from ballistically acceptable lots may be discarded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.

If for any reason the proving ground personnel consider that the test conditions have detrimentally affected the test result, the test result shall be declared invalid and a new test shall be performed on additional sample cartridges."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (modifications, additions, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-0565)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:50Z
Check the source to verify that this is the current version before use.